

Oracle EPM Predictive Cash Forecasting

Direct Cash Flow Method

Advanced Lab Guide | Product Design + Implementation + Training + Technical Reference +
Troubleshooting

Advanced-stage focus

Design a direct-method cash forecast that converts source transactions and operating assumptions into explicit cash receipts and payments, then turns those flows into actionable liquidity decisions.

Oracle-inspired conceptual UI used for training and solution-design purposes. It is not a reproduction of Oracle proprietary screens.

1. Purpose and Learning Outcomes

Predictive Cash Forecasting is an Oracle EPM Planning application type built around the direct cash flow method. Instead of deriving cash indirectly from accrual P&L and balance-sheet movements, the direct method forecasts the timing and amount of specific cash receipts and cash payments. The output is a transparent liquidity bridge that treasury can reconcile to ERP transactions and bank activity.

What makes this direct-method design valuable

Cash managers can explain exactly why cash moves: customer collections, supplier payments, payroll, taxes, capex, debt service, financing proceeds, investments, and other explicit categories. That improves traceability, accountability, and actionability.

Learning outcomes

- Design a direct-method cash forecast model for daily, weekly, or monthly horizons.
- Map ERP and bank data to operating, investing, and financing cash line items.
- Configure forecast methods by line item, entity, and forecast period.
- Build a 13-week rolling forecast from opening cash through funding actions.
- Measure forecast accuracy, bias, and Forecast Value Add (FVA).
- Diagnose integration, mapping, calendar, method-selection, and calculation failures.
- Define governance, workflow, security, audit, and production support controls.

Intended audience

Role	Why this guide matters
Oracle EPM Consultant	Configuration, data integration, rules, forms, deployment
Treasury / Cash Manager	Liquidity decisions, assumptions, exception management
Controller / FP&A	Driver alignment, forecast governance, scenario analysis
Solution Architect	Enterprise design, security, automation, operating model
Advanced Student	Hands-on lab with realistic calculations and troubleshooting

2. Direct Cash Flow Method - Conceptual Model

Direct Cash Flow Method - Reference Architecture

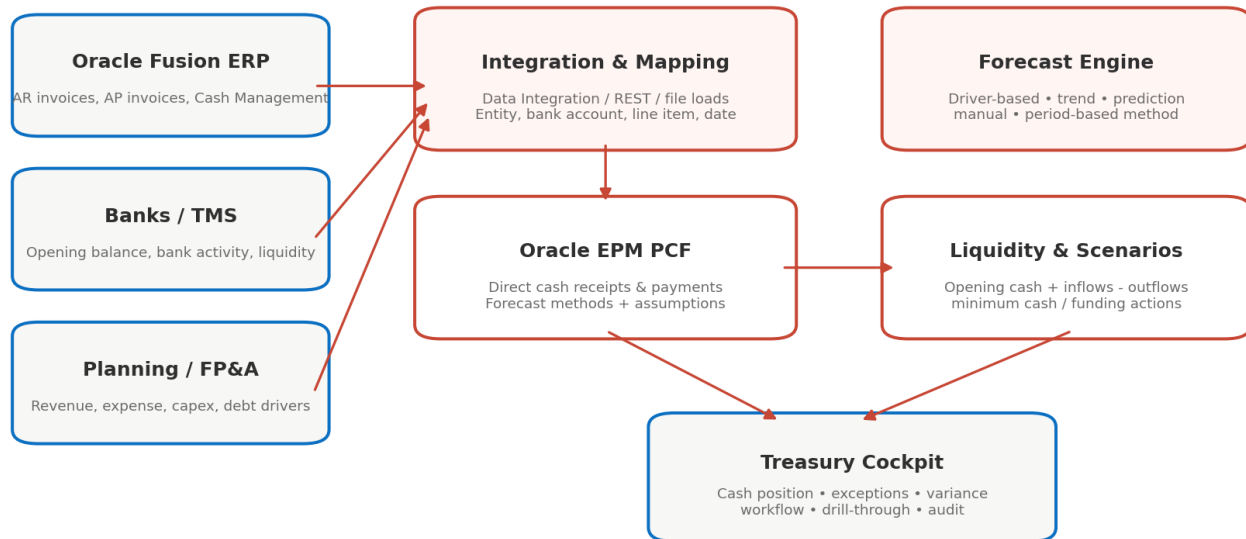


Figure 1. Reference architecture for a direct-method cash forecast.

Core formula

Liquidity equation

Ending Cash = Opening Cash + Cash Inflows - Cash Outflows + Financing / Investment Actions

The key distinction is that each line item is a forecasted cash event, not merely an accounting expense or revenue. The model therefore depends on timing logic - invoice due dates, payment terms, DSO/DPO, payroll calendars, tax schedules, project milestones, debt schedules, and treasury overrides.

Cash-flow hierarchy

Category	Typical Direct Cash Line Items	Primary Source
Operating inflows	Customer collections, service receipts, rebates	AR / order-to-cash / revenue drivers
Operating outflows	Supplier payments, payroll, taxes, rent, utilities	AP / payroll / tax schedules
Investing	Capex payments, asset-sale proceeds, investments	Project plans / fixed assets / treasury
Financing	Debt drawdowns, repayments, interest, dividends	Debt schedules / treasury / board-approved plans
Cash balances	Opening cash, restricted cash, bank balances	Cash Management / bank feeds

3. End-to-End Direct Forecast Process

Direct Method Forecast Process

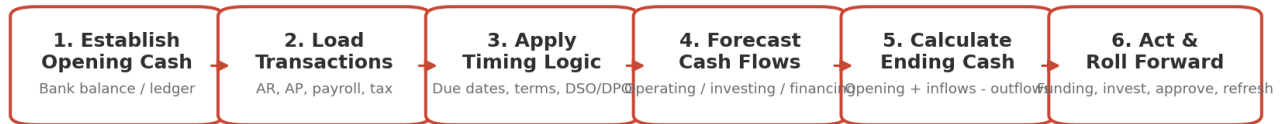


Figure 2. Direct-method forecasting process from opening cash through liquidity action.

Process design notes

Stage	Design Question	Control
Opening cash	What is the authoritative cash balance and timestamp?	Bank-to-ledger reconciliation complete
Transactions	Which open AR/AP items are in scope?	Load completeness and duplicate checks
Timing	How are due dates, terms, and holidays treated?	Calendar + assumption validation
Forecast	Which method is used by line item and horizon?	Method assignment by entity / period
Liquidity	What minimum cash buffer applies?	Threshold and headroom rules
Action	Who may override or trigger funding?	Approval, comments, audit trail

Recommended navigation concept

Oracle-inspired navigation

Home > Periodic Cash Forecast > Overview > Assumptions > Drivers - Cash Inflow > Drivers - Cash Outflow > Rolling Forecast > Predicted > Variance & Accuracy > Exceptions > Administration

4. Data Architecture and Mapping

Data-Flow Diagram - Transaction to Cash Forecast

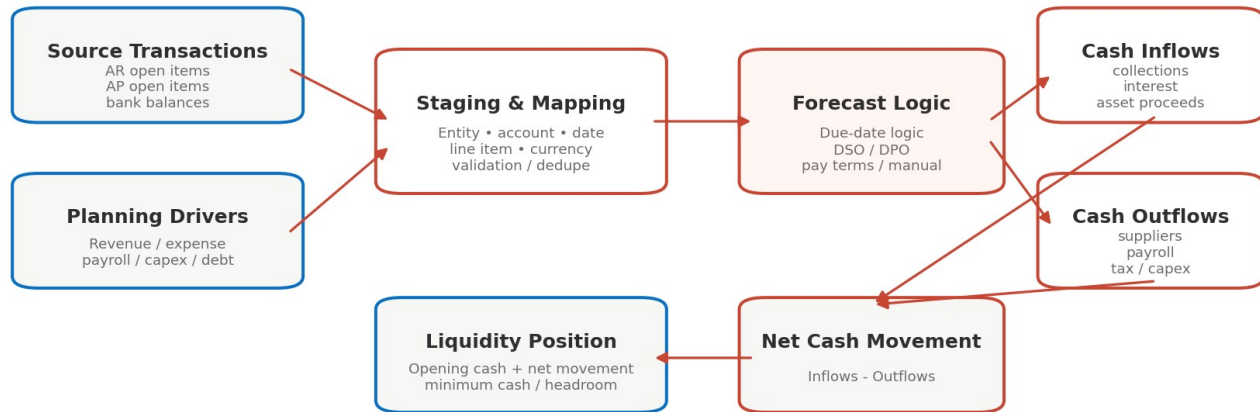


Figure 3. Direct-method data flow from transaction sources into liquidity position.

Minimum mapping dimensions

Dimension / Attribute	Example	Purpose
Entity	US01	Legal-entity ownership and security
Cash Line Item	Customer Collections	Direct receipt / payment classification
Bank Account	BOFA_US_001	Opening balance and cash-position reconciliation
Date / Period	2026-W35	Forecast timing and roll-forward
Currency	USD	Local / reporting currency handling
Version	Working / Approved	Workflow and scenario governance
Source	AR / AP / Manual	Lineage and troubleshooting
Customer / Vendor / Project	Optional custom dimension	Granular driver analysis

Mapping example

Source Record	Source Amount	Forecast Line Item	Forecast Week	Rule
AR INV-10081	\$1.20M	Customer Collections	W2	Due date + customer behavior
AP INV-50442	\$0.65M	Supplier Payments	W1	Payment date / DPO
Payroll Batch	\$1.10M	Payroll	W2	Payroll calendar
Project CAPEX-27	\$0.80M	Capital Expenditure	W5	Milestone + payment terms
Debt Schedule	\$0.40M	Debt Repayment	W8	Contractual due date

5. Forecast Method Selection

Forecast Method Decision Tree by Cash Line Item

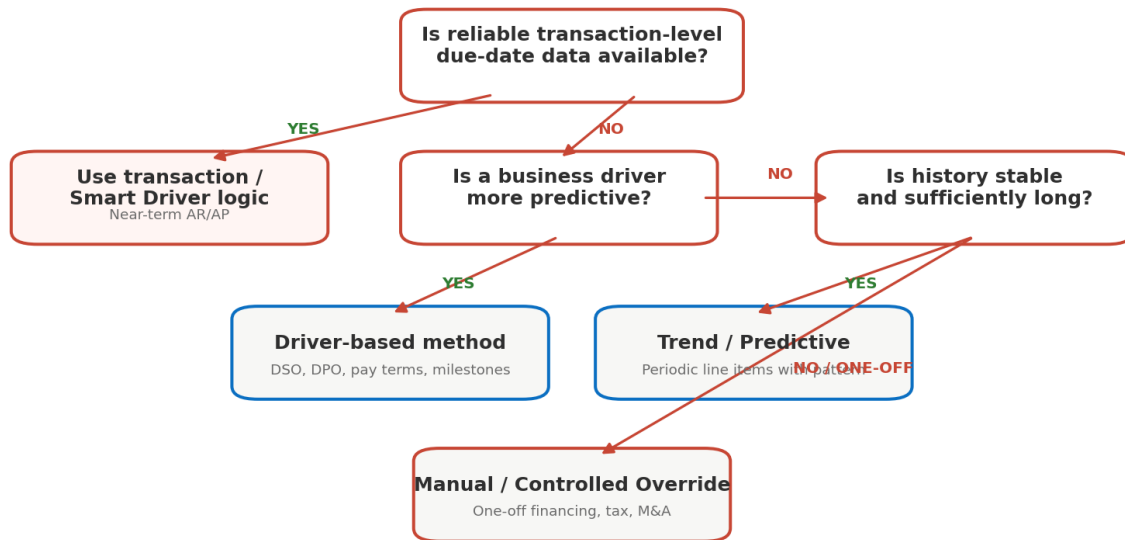


Figure 4. Method-selection decision tree for direct cash line items.

Oracle PCF supports multiple forecast approaches by line item and entity, and period-based methods can use different approaches for near-term and farther horizons. The direct-method design should favor the most granular reliable source available, then fall back to drivers, trend/predictive methods, or controlled manual input.

Method pattern library

Line Item	Near-Term Method	Farther-Term Method	Why
Customer collections	Open AR / Smart Driver	DSO / revenue driver	Transaction precision near term; planning driver farther out
Supplier payments	Open AP / Smart Driver	DPO / expense driver	Due-date precision + operational extension
Payroll	Calendar / driver	Driver	Highly deterministic recurring schedule
Tax	Manual / tax calendar	Rate / driver	Event-driven and jurisdiction-specific
Capex	Project milestone	Capital plan / manual	Milestone timing is more meaningful than trend
Interest / debt	Debt schedule	Driver / manual	Contractual cash timing
Other operating cash	Transaction if available	Trend / predictive	Use stable historical pattern only where appropriate

6. Oracle-Inspired Configuration Mockup

Oracle EPM | Predictive Cash Forecasting

Home > Periodic Cash Forecast > Assumptions > Direct Method

Entity
US01

Forecast Range
13 Weeks

Currency
USD

Version
Working

Forecast Method Assignment

Cash Line Item	Method 1	End Period	Method 2	Driver / Source
Customer Collections	Smart Driver	W4	DSO Driver	AR Open Items
Supplier Payments	Smart Driver	W4	DPO Driver	AP Open Items
Payroll	Driver Based	W13	-	Payroll Calendar
Tax Payments	Manual	W13	-	Tax Schedule
Capex	Milestone	W13	-	Project Plan
Interest / Debt	Driver Based	W13	-	Debt Schedule

Save & Calculate

Figure 5. Conceptual forecast-method configuration panel.

Configuration checklist

- Create or enable the Predictive Cash Forecasting application artifacts.
- Define daily or periodic forecasting, fiscal calendar, entities, currencies, and forecast horizon.
- Review and extend cash line items under operating, investing, and financing structures.
- Set default methods for every required entity and line item.
- Configure near-term and farther-period method boundaries where appropriate.
- Load driver assumptions: pay terms, DSO/DPO, due dates, milestone percentages, schedules.
- Validate user preferences, POVs, workflow, and security before end-user testing.

Advanced-stage control

Do not assign one forecast method globally without testing line-item behavior. Method quality should be evaluated by horizon, source maturity, volatility, forecast error, and explainability.

7. Advanced Lab Scenario - 13-Week Direct Cash Forecast

Scenario: US01 is a manufacturing entity forecasting USD liquidity over 13 weeks. Treasury requires a minimum cash buffer of \$10.0M. The forecast combines open-item transaction logic for near-term AR/AP and driver-based methods for the remaining horizon.

Lab objectives

- Calculate explicit weekly cash receipts and payments.
- Identify the lowest liquidity point and required funding action.
- Apply a stress scenario to collections and supplier payments.
- Create an approved treasury override with audit commentary.
- Reconcile forecast and actual cash and calculate forecast accuracy.

Starting assumptions

Assumption	Value
Opening cash at start of W1	\$12.0M
Minimum operating cash buffer	\$10.0M
Forecast horizon	13 weeks
Near-term transaction window	W1-W4
Farther-horizon driver window	W5-W13
Stress scenario	Collections -10%; supplier payments +5% in W5-W8
Funding instrument	Revolver; draw only if ending cash < \$10.0M

8. Base Forecast - Worked Numerical Example

Week	Opening Cash	Inflows	Outflows	Net Move	Ending Cash	Liquidity
W1	12.0	8.2	7.5	+0.7	12.7	OK
W2	12.7	7.8	8.1	-0.3	12.4	OK
W3	12.4	8.7	8.0	+0.7	13.1	OK
W4	13.1	9.1	8.9	+0.2	13.3	OK
W5	13.3	7.9	9.4	-1.5	11.8	OK
W6	11.8	8.5	8.2	+0.3	12.1	OK
W7	12.1	9.4	8.8	+0.6	12.7	OK
W8	12.7	8.8	9.3	-0.5	12.2	OK
W9	12.2	9.6	8.7	+0.9	13.1	OK
W10	13.1	9.1	9.5	-0.4	12.7	OK
W11	12.7	10.0	9.2	+0.8	13.5	OK
W12	13.5	9.7	10.1	-0.4	13.1	OK
W13	13.1	10.3	9.6	+0.7	13.8	OK

Example calculation - Week 5

Opening Cash W5 = \$12.1M. Forecast inflows = \$7.9M. Forecast outflows = \$9.4M. Therefore Net Cash Movement = \$7.9M - \$9.4M = -\$1.5M. Ending Cash = \$12.1M - \$1.5M = \$10.6M. Because \$10.6M remains above the \$10.0M buffer, no revolver draw is required in the base case.

Interpretation

The base case stays above the minimum buffer. The lowest forecast cash position is \$11.8M, so treasury has limited but positive headroom.

9. Build Cash Inflows - Customer Collections

Near-term transaction method

For W1-W4, use open AR items and their expected payment dates. Adjust where customer behavior differs from contractual terms. Example: an invoice due in W2 may be shifted to W3 for a customer whose actual payment history consistently runs seven days late.

Farther-term DSO method

For W5-W13, forecast revenue can be translated into cash collections using DSO or payment-term assumptions. A simplified weekly approximation is to shift forecast revenue into future cash periods based on the average collection delay.

Driver	Example
Forecast revenue in W5	\$10.0M
Expected cash conversion ratio	85% within modeled horizon
Average DSO	35 days
Approximate timing	Majority collected about 5 weeks later
Direct cash line item	Customer Collections

Student task

- Load or enter AR open-item data for W1-W4.
- Configure DSO assumptions for W5-W13.
- Run the appropriate cash-inflow calculation rule.
- Confirm calculated receipts populate the rolling forecast in the expected weeks.
- Drill back to source detail for one material receipt.

10. Build Cash Outflows - Supplier Payments

Near-term AP logic

Use open AP invoices, payment dates, and payment terms for the near-term period. The forecast should respect planned payment dates, terms, and any treasury payment holds that are governed and documented.

Farther-term DPO logic

For farther periods, expense forecasts can be converted into supplier payments using DPO assumptions. Where multiple pay terms exist, allocate the expense across the relevant timing percentages and forecast periods.

Expense Driver	Amount	Pay Term Assumption	Direct Cash Result
Raw materials	\$4.0M	50% at 30 days; 50% at 45 days	\$2.0M shifted ~4 weeks; \$2.0M shifted ~6 weeks
Logistics	\$1.2M	NET30	\$1.2M shifted ~4 weeks
Professional services	\$0.6M	NET15	\$0.6M shifted ~2 weeks

Control point

Avoid double counting

If open AP invoices are already forecast through transaction-level logic, do not also forecast the same obligations through expense-driver logic. Establish a clear actual/open-item/driver cutover by horizon.

11. Payroll, Tax, Capex, and Financing

Cash Flow	Recommended Direct-Method Design	Example Control
Payroll	Payroll calendar or employee-cost driver by pay date	Tie to payroll register / headcount plan
Tax payments	Jurisdiction tax calendar, estimate schedule, manual controlled input	Tax manager approval + evidence
Capex	Project milestones + payment terms	Project ID and approved capital plan
Debt repayment	Contractual maturity schedule	Treasury-owned debt master
Interest	Rate + principal / contractual schedule	Reconcile to treasury debt system
Dividends	Board-approved dates and amounts	Approval evidence required
Investments	Treasury action / policy-driven	Liquidity threshold and counterparty limits

Direct-method design principle

Every material line item should answer four questions: What cash event is occurring? How much? When will it occur? What source or approved assumption supports the amount and timing?

12. Stress Scenario and Liquidity Decision

Week	Stress Inflows	Stress Outflows	Ending Cash	Headroom vs \$10M
W1	8.20	7.50	12.70	2.70
W2	7.80	8.10	12.40	2.40
W3	8.70	8.00	13.10	3.10
W4	9.10	8.90	13.30	3.30
W5	7.11	9.87	10.54	0.54
W6	7.65	8.61	9.58	-0.42
W7	8.46	9.24	8.80	-1.20
W8	7.92	9.77	6.96	-3.04
W9	9.60	8.70	7.86	-2.14
W10	9.10	9.50	7.46	-2.54
W11	10.00	9.20	8.26	-1.74
W12	9.70	10.10	7.86	-2.14
W13	10.30	9.60	8.56	-1.44

Decision rule

Funding trigger

If ending cash falls below \$10.0M, treasury should evaluate a revolver draw, payment rescheduling, discretionary capex deferral, or other approved liquidity action. In this scenario, the lowest cash is \$6.96M in W8.

Student exercise: Create a controlled \$1.0M revolver draw in the first week where the stress case would otherwise breach the buffer. Recalculate and document the reason, approver, source evidence, and forecast version.

13. Rolling Forecast and Actual Cutover

A direct-method forecast becomes most useful when it is continuously rolled. When a week closes, actual bank and ERP cash activity replaces forecast for that period, the forecast error is calculated, and a new future week enters the horizon.

Before Roll	After Roll
W1-W13 forecast	W1 becomes actual; W2-W14 becomes active forecast
Opening cash estimated	Opening cash refreshed from reconciled bank balance
Forecast transactions	Closed-period actual transactions retained for analysis
W13 horizon end	New W14 added using assigned forecast method
Working version	Updated working version moves through review / approval

Roll-forward controls

- Actual-period lock prevents accidental overwrite.
- Opening cash for the new horizon is sourced from the reconciled closing balance.
- Overrides are reviewed: expire, carry, or reapprove according to policy.
- New-period driver and source data are loaded before calculation.
- Forecast-method boundaries advance with the horizon.

14. Forecast Accuracy, Bias, and FVA

Key measures

Metric	Formula	Interpretation
Absolute Error	Forecast - Actual	Magnitude of miss
APE	F-A / Actual	Percent error for a line item / period
MAPE	Average APE	Overall accuracy indicator
Bias	Average (Forecast - Actual)	Systematic over / under forecasting
FVA	Baseline error - Process-step error	Whether a forecast step improves accuracy

Worked variance example

Line Item	Forecast	Actual	Variance	APE
Customer Collections	\$8.20M	\$7.75M	-\$0.45M	5.8%
Supplier Payments	\$7.50M	\$7.90M	+\$0.40M	5.1%
Payroll	\$1.10M	\$1.08M	-\$0.02M	1.9%
Tax	\$0.45M	\$0.60M	+\$0.15M	25.0%

Advanced analysis

A high-error line item should trigger root-cause analysis before changing the forecast method. Possible causes include stale source data, mapping gaps, unusual one-offs, calendar timing, assumption drift, or inappropriate method selection.

15. Cash Manager Dashboard Mockup

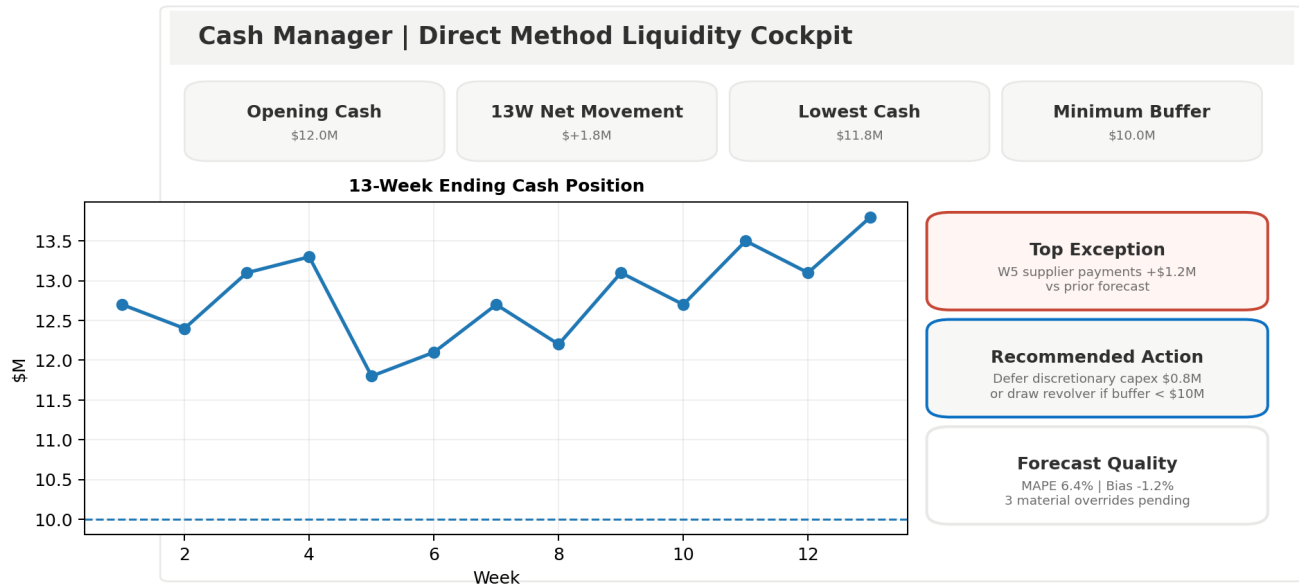


Figure 6. Conceptual direct-method liquidity cockpit for treasury.

Dashboard design requirements

- Opening, ending, and minimum cash by entity / currency.
- Cash inflows and outflows by operating, investing, and financing categories.
- Lowest-cash date / week and headroom to minimum buffer.
- Base vs stress scenario bridge.
- Top forecast changes since prior forecast.
- Forecast accuracy / bias by line item and horizon.
- Override count, workflow status, and exception aging.
- Drill-through link or reference to ERP / source transaction where available.

16. Workflow, Overrides, and Governance

Stage	Owner	Allowed Action	Required Evidence
Prepare	Cash Analyst	Run forecast, enter permitted overrides	Source reference + comment
Review	Cash Manager	Challenge assumptions, approve / reject	Variance rationale
Approve	Treasury Controller	Approve material overrides / funding	Approval and policy check
Publish	Treasury Lead	Release approved forecast	Version stamp / date
Audit	Administrator / Internal Audit	Review lineage and changes	Audit trail / job logs

Override policy example

Materiality control

Overrides greater than \$0.5M or 10% of the original line-item forecast require a reason code, evidence attachment or reference, and reviewer approval. Overrides should be time-bound and revalidated after each roll.

17. Security and Segregation of Duties

Role	Access
Service Administrator	Application configuration, metadata, integration, security
Cash Forecast Administrator	Methods, assumptions, rules, forms, jobs
Cash Manager	Entity-scoped assumptions, review, exceptions, controlled overrides
Cash Analyst / Preparer	Load / input allowed data, run assigned forecasts
Controller / Reviewer	Review, approve, analyze variance
Read Only / Executive	Dashboards and approved forecast only

Security design tests

- US01 preparer cannot access EU01 unless explicitly granted.
- Preparer cannot approve their own material override.
- Bank-account data is limited to authorized entity / role scope.
- Administrative forms and integration credentials are not exposed to end users.
- Approved versions are protected from unintended writeback.

18. Automation and Production Runbook

Recommended daily / weekly job chain

Seq	Job	Expected Result
1	Load bank balances	Opening cash refreshed and reconciled
2	Load AR / AP open items	Near-term transactional drivers current
3	Load planning drivers / schedules	Farther-horizon drivers available
4	Run validation and mapping checks	No blocking exceptions
5	Calculate cash inflows / outflows	Direct line items populated
6	Calculate rolling forecast	Ending cash and liquidity metrics updated
7	Run variance / exception rules	Material changes surfaced
8	Publish dashboard / notify reviewers	Users review working forecast

Operational SLA example

Control	Target
Source load completion	Before 06:15 local treasury time
Blocking mapping errors	0
Forecast calculation	< 10 minutes for agreed scope
Dashboard available	By 06:45
Material exception review	By 09:30
Approved forecast cut-off	Per treasury calendar

19. Troubleshooting Handbook

Job Console & Error Diagnostics

Time	Job	Entity	Status	Message / Root Cause
06:00	Load AR Open Items	US01	SUCCESS	42,810 rows loaded; 0 rejects
06:07	Load AP Open Items	US01	WARNING	18 rows missing payment terms; defaulted to NET30
06:12	Map Bank Balances	US01	ERROR	Bank account 001-984 not mapped to Cash Account
06:20	Calculate Forecast	US01	BLOCKED	Upstream bank mapping error
06:31	Calculate Forecast	US01	SUCCESS	13 weeks calculated after mapping correction

Figure 7. Example job console and root-cause diagnostics.

Symptom	Likely Cause	Diagnostic	Resolution
Opening cash is zero	Bank mapping / load issue	Check bank account mapping and latest balance load	Map account; reload; reconcile
Collections shifted to wrong week	Due date / calendar / DSO issue	Inspect source due date and assumption	Correct calendar or assumption; recalc
Double-counted supplier payments	Open AP + expense driver overlap	Compare source lineage by period	Define cutover and exclude overlapping driver amount
Forecast does not update after assumption change	Rule not executed / stale form	Check job console and calc timestamp	Run appropriate rule / refresh
Entity missing from form	Security / metadata issue	Check entity assignment and valid intersections	Grant access / correct metadata
Large unexplained variance	One-off actual, stale method, source change	Drill to transactions and prior assumptions	Classify cause; update method only if persistent
Manual override disappears after roll	Override not configured to carry	Review override governance and new horizon	Re-enter / reapprove per policy

20. Product Design Requirements

Capability	Functional Requirement	Acceptance Criterion
Direct cash hierarchy	Operating, investing, financing inflows/outflows	All material flows map to one transparent line item
Method assignment	Method by entity, line item, horizon	Near/far method boundaries calculate correctly
Transaction lineage	Link forecast to source reference	Material forecast items trace to source or assumption
Rolling horizon	Actual drops in; new forecast period added	No gap or duplicated period after roll
Scenario planning	Base / stress / management case	Scenario changes isolated and reproducible
Liquidity controls	Minimum cash and funding action	Breach produces exception and action workflow
Governance	Override approval / audit	Every material override has author, reason, date, status
Observability	Logs and diagnostics	Blocking errors are identifiable without database access

21. Implementation Checklist

- Confirm business purpose and forecast horizon.
- Define cash line-item hierarchy and materiality.
- Document authoritative opening-cash source.
- Inventory AR, AP, Cash Management, payroll, tax, capex, debt, and planning sources.
- Design mappings and custom dimensions.
- Configure forecast methods by entity / line item / period.
- Load assumptions and establish transaction/driver cutover.
- Configure forms, navigation, dashboards, and user preferences.
- Implement security and workflow.
- Build integration and calculation job chain.
- Define reconciliation, accuracy, FVA, and exception controls.
- Complete SIT with reconciliation to source.
- Complete UAT with treasury scenarios.
- Document production runbook and support ownership.
- Establish rolling forecast governance and continuous improvement cadence.

22. Advanced Student Validation Exercises

Exercise	Expected Evidence
1. Forecast build	13-week direct-method forecast with opening cash, inflows, outflows, ending cash
2. AR drill-through	One collection traced to source invoice / driver
3. AP cutover test	Proof that open-item and driver logic do not double count
4. Stress scenario	Collections -10%, supplier payments +5%, liquidity impact quantified
5. Funding action	Revolver draw or capex deferral triggered by buffer rule
6. Accuracy test	APE / MAPE and bias calculated for at least four line items
7. Override governance	Material override with reason, approver, and version
8. Error recovery	Mapping error reproduced, diagnosed, fixed, and rerun
9. Roll-forward	Closed week converted to actual and new week added
10. Design defense	Explain why each line item uses its chosen forecast method

Completion standard

The lab is complete only when the student can reconcile the final cash position to opening cash plus explicit direct cash movements and explain every material adjustment or exception.

23. Technical Reference - Oracle PCF Concepts

Key Oracle concepts reflected in this guide include: Predictive Cash Forecasting as a Planning application type; daily, weekly, and monthly rolling forecasts; direct cash flow methodology; operational, investing, and financing cash line items; role-based navigation; multiple forecast methods including driver-based, trend-based, Predictive Planning, manual input, and period-based method selection; and integration with Oracle Fusion Cloud ERP modules such as Accounts Receivable, Accounts Payable, and Cash Management.

Reference links

Oracle Documentation	URL
Predictive Cash Forecasting Overview	https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/cashu/cash_overview.html
Creating Predictive Cash Forecasting Applications	https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/pcf-create-applications/index.html
Forecast Methods and Driver Methods	https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/pcf-forecast-methods/index.html
Using Predictive Planning for Cash Forecasting	https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/cashu/cash_predictions.html
Overview of Predictive Cash Forecasting with ERP	https://docs.oracle.com/en/cloud/saas/financials/26b/fapp/overview-of-predictive-cash-forecasting.html

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